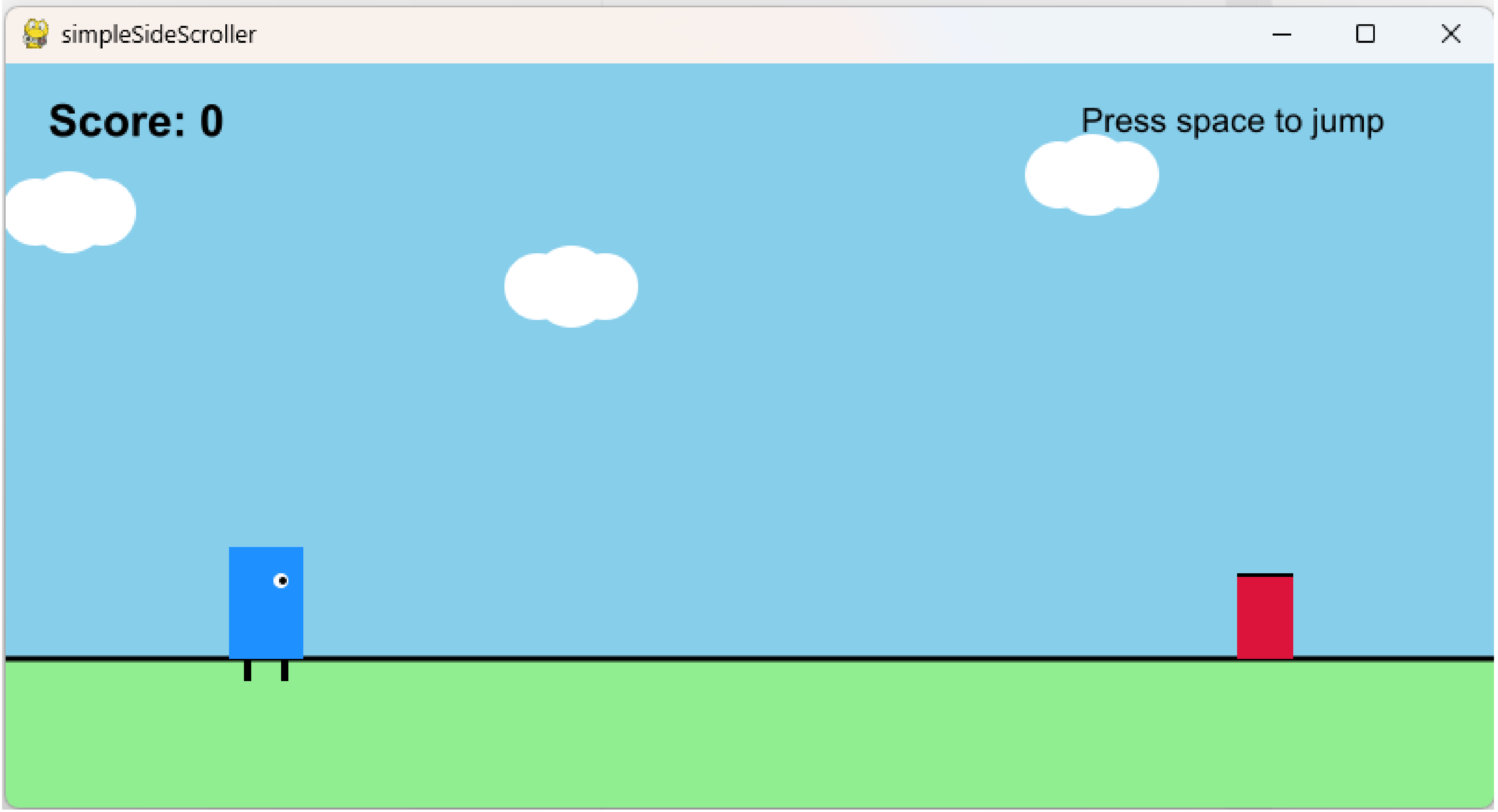
OOPy Animation
Side Scroller Game

Hend Gedawy

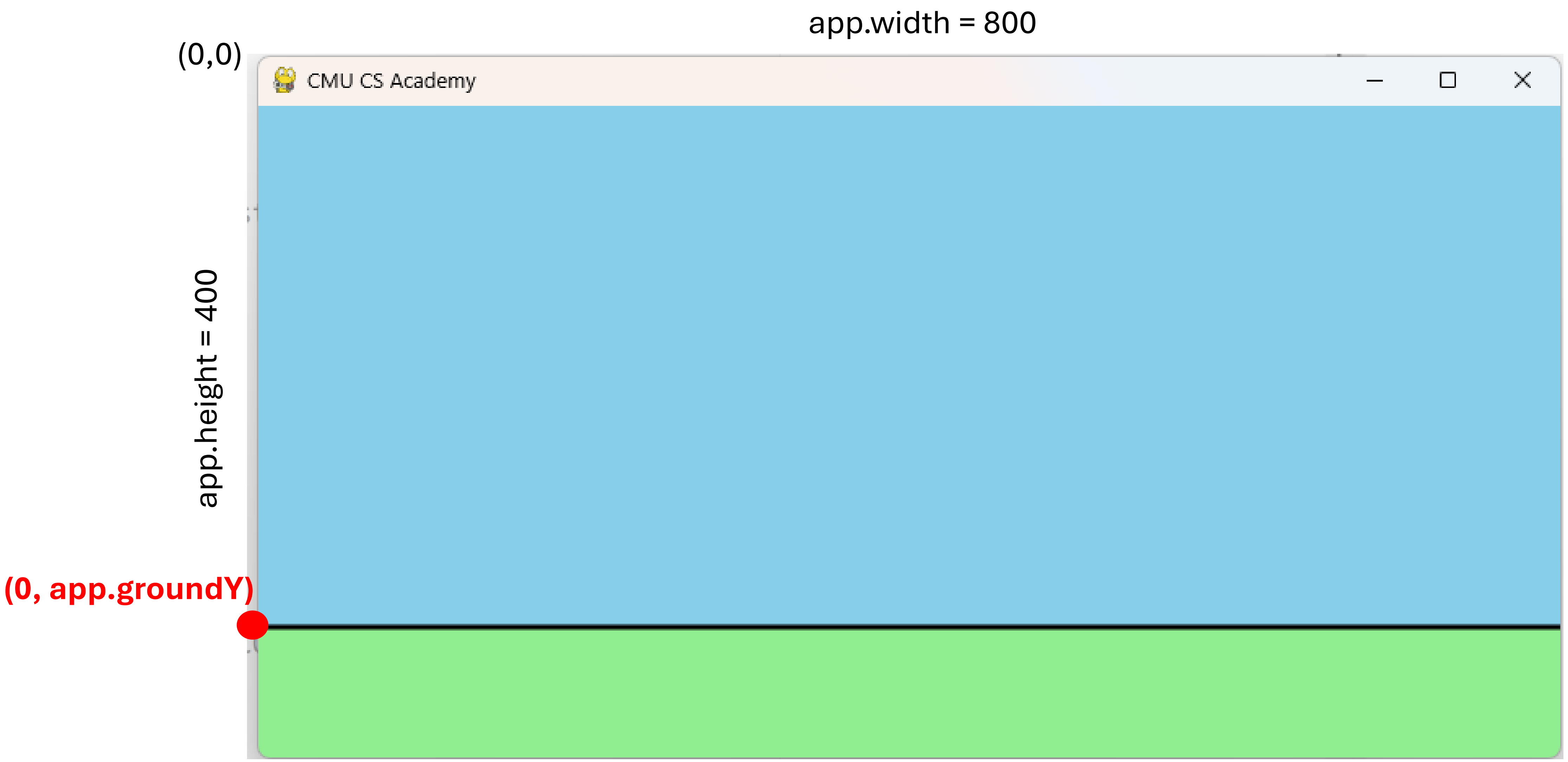
Entities?



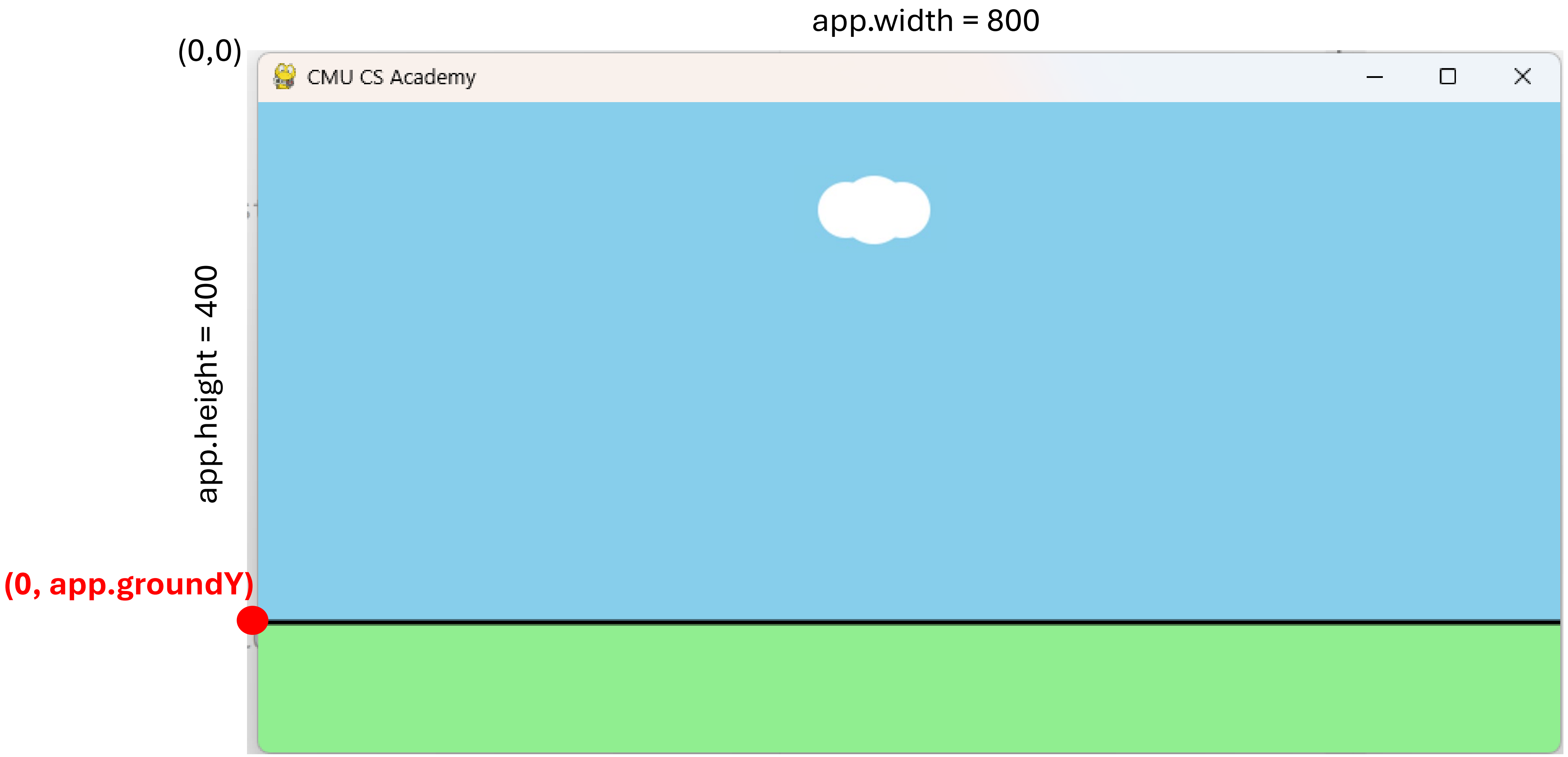
Entities?


simpleSideScroller
Score: 0
Game Over!
Final Score: 0
Press r to restart

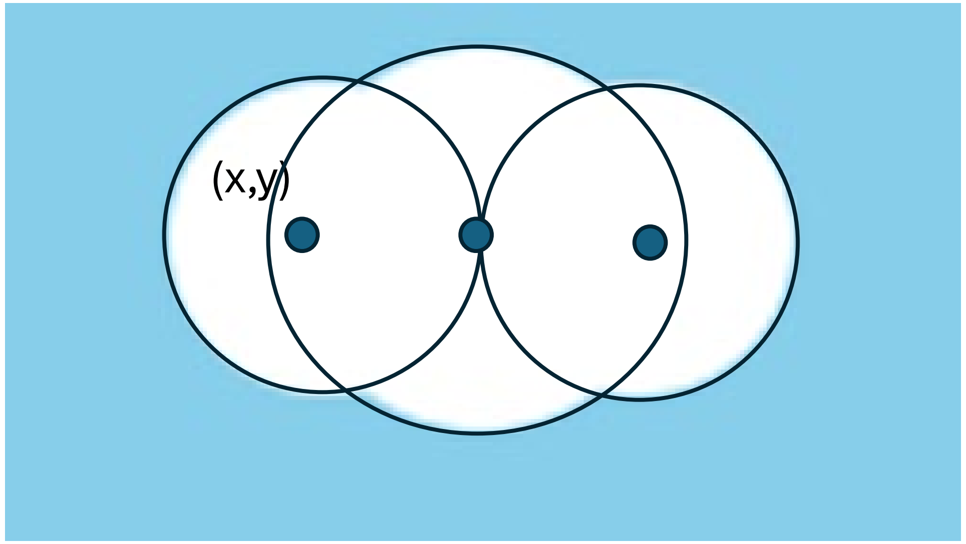
Background: Sky And Ground



Background: Clouds



Background: Drawing a Cloud



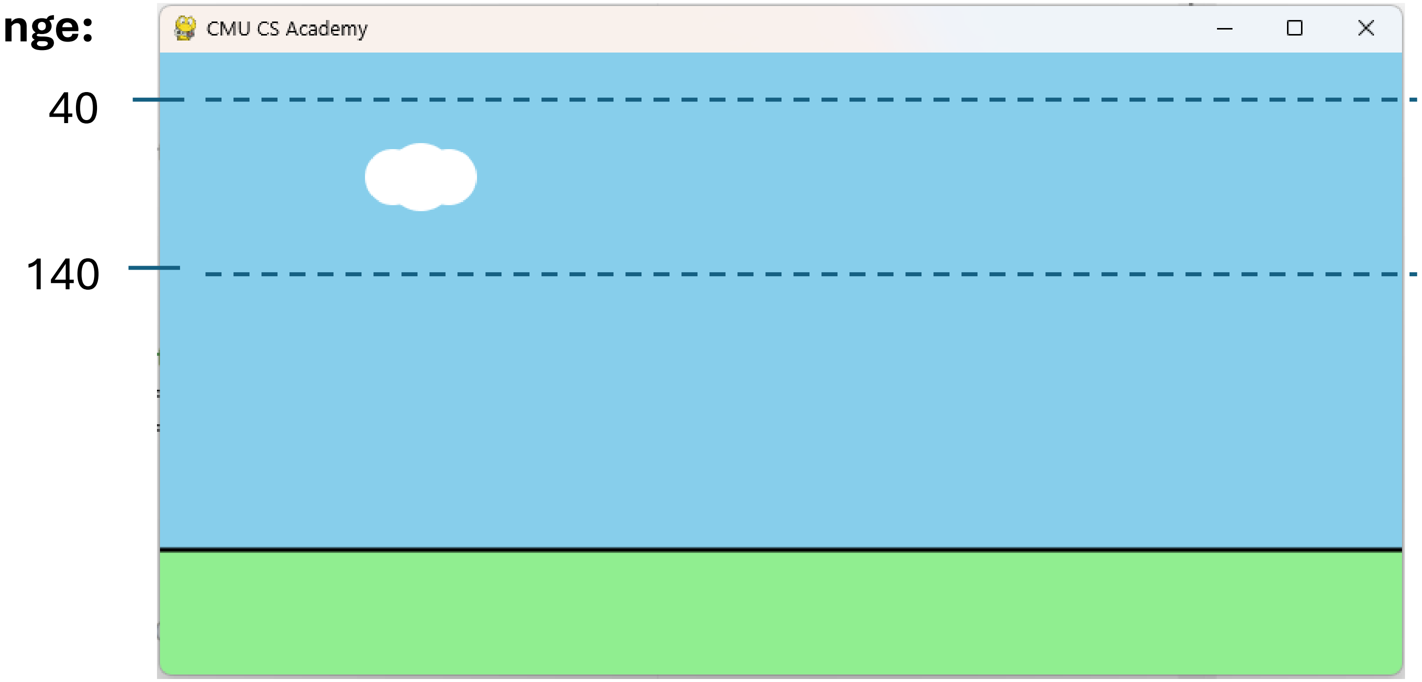
Small circles radius = 18

Large circle radius = 22

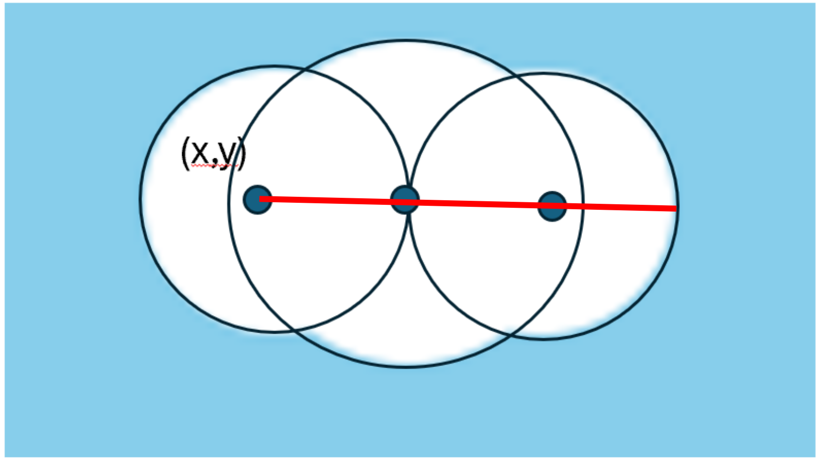
```
drawCircle(self.x, self.y, 18, fill='white')
drawCircle(self.x + 18, self.y, 22, fill='white')
drawCircle(self.x + 36, self.y, 18, fill='white')
```

Background: Cloud Motion

Clouds y Value range:



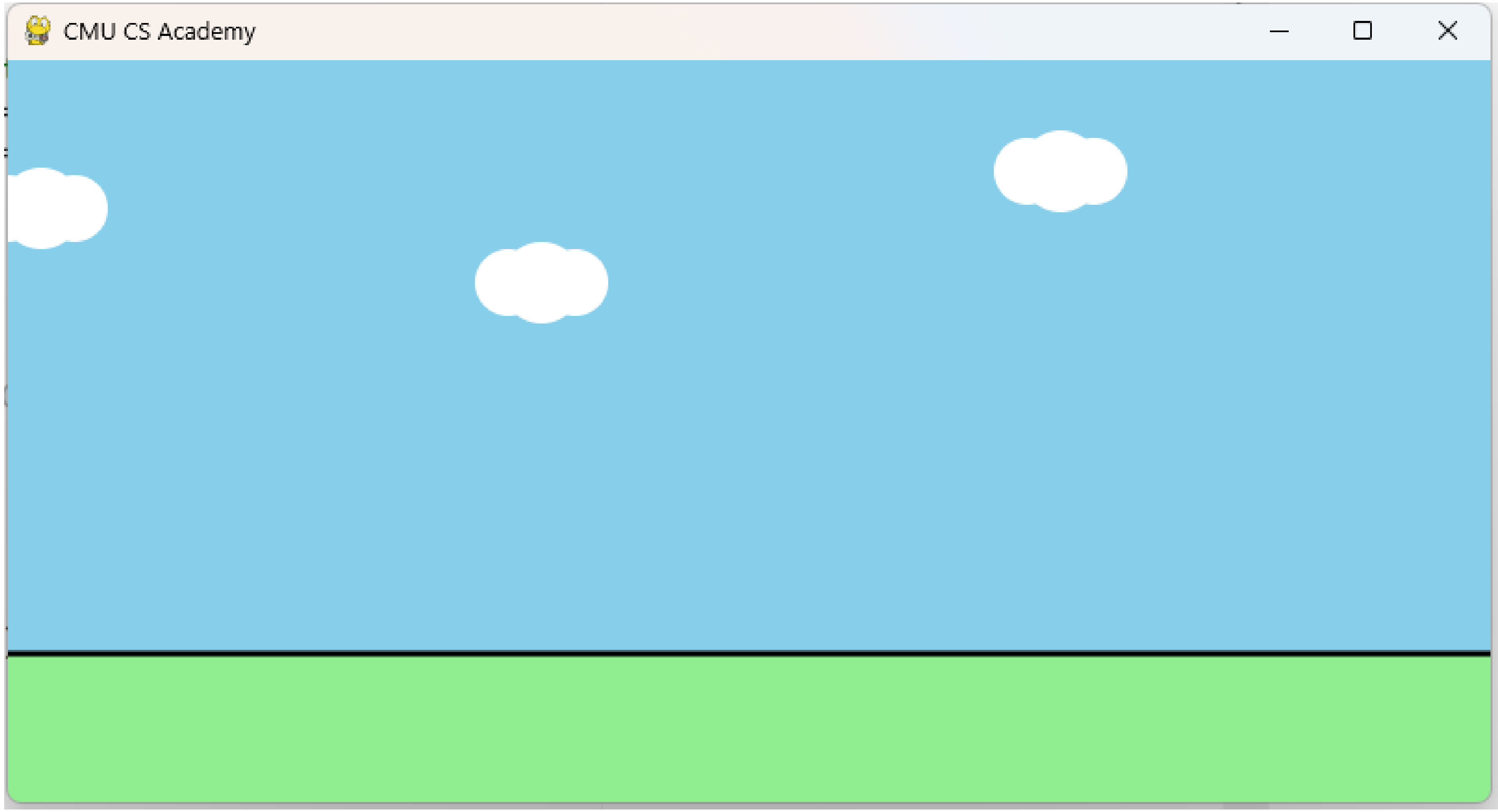
Small circles radius = 18



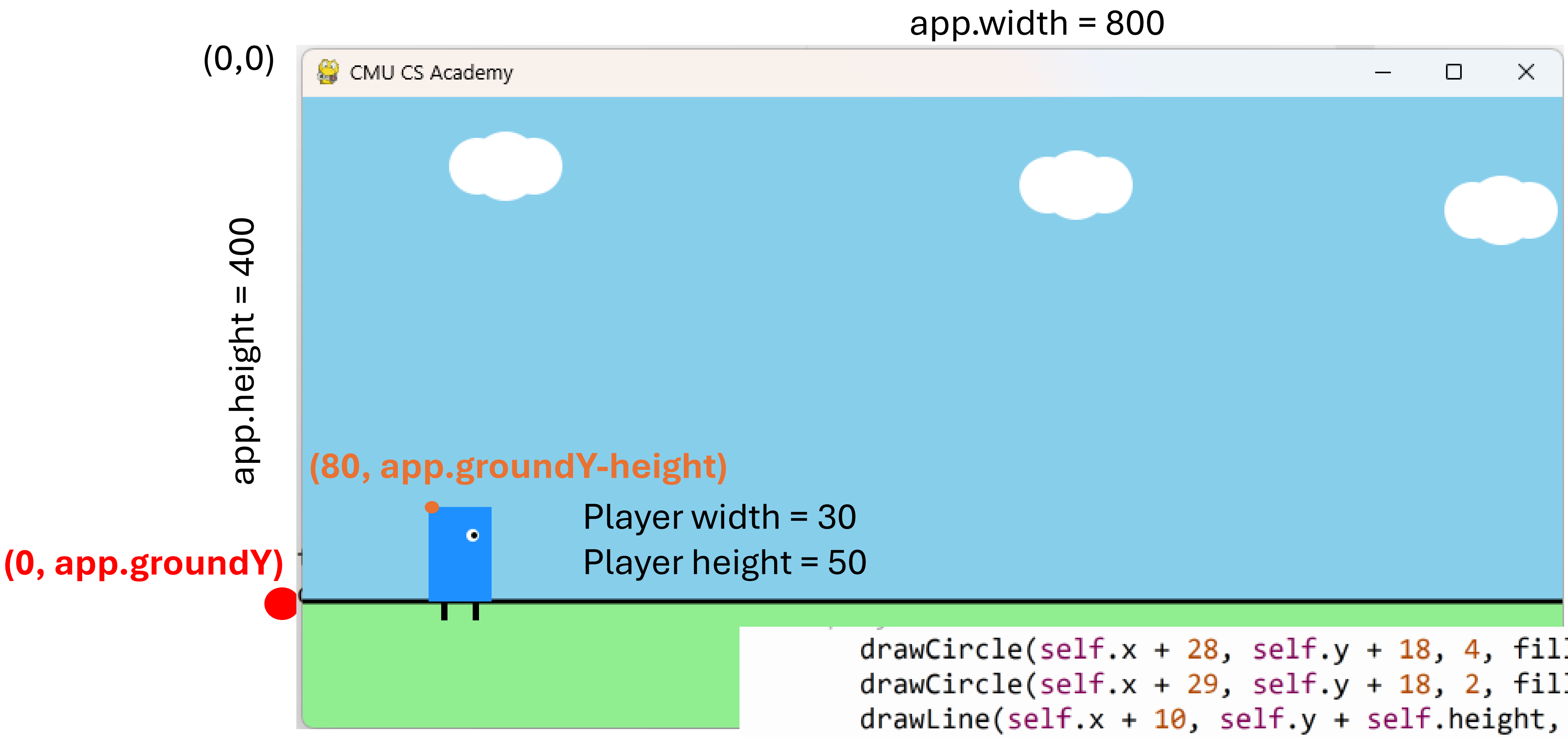
This line (from x to the end of the cloud) = 18+18+18 = 54

For the whole cloud to get off screen, x < -54

Background: Multiple Clouds

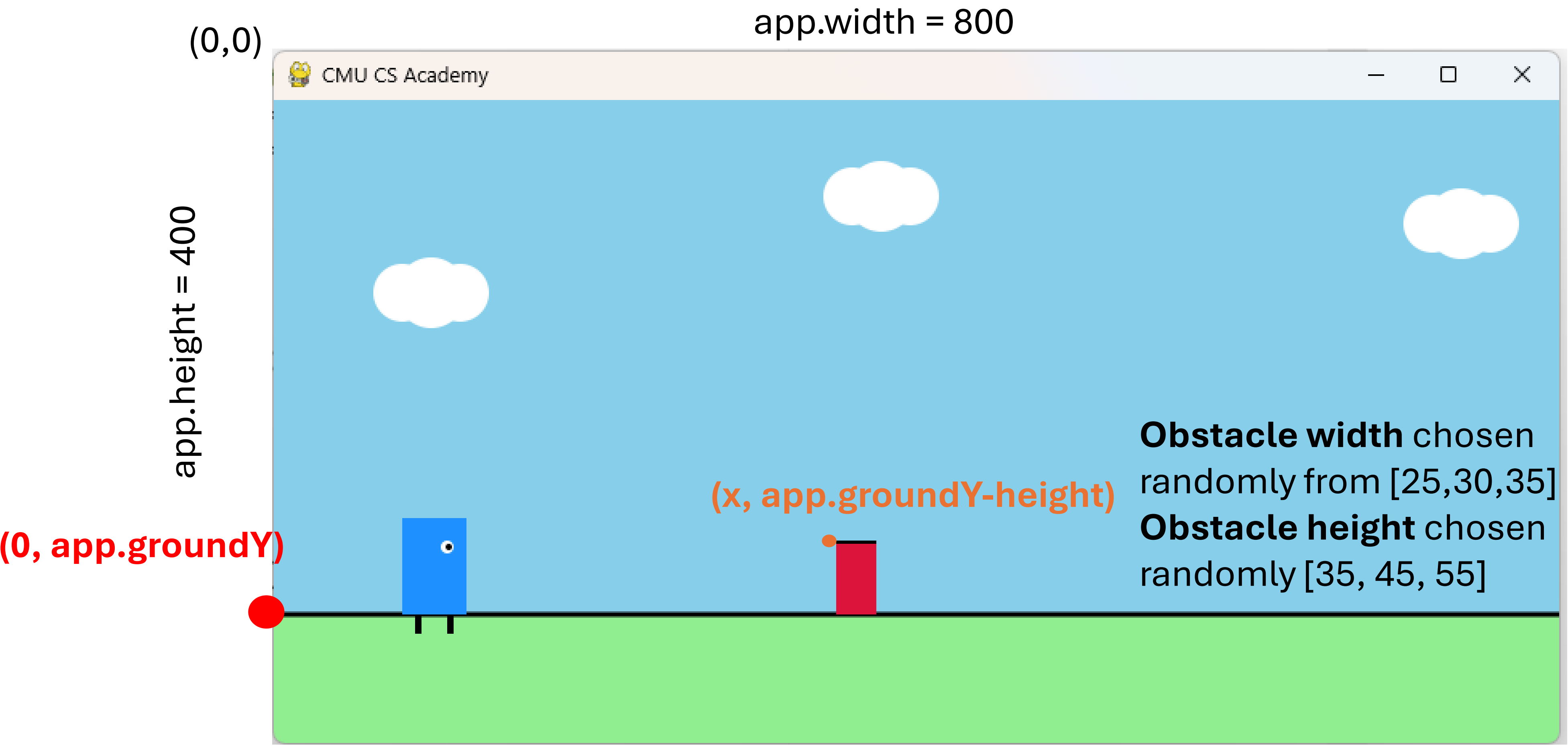


Player: Drawing




```
drawCircle(self.x + 28, self.y + 18, 4, fill='white')
drawCircle(self.x + 29, self.y + 18, 2, fill='black')
drawLine(self.x + 10, self.y + self.height,
         self.x + 10, self.y + self.height + 12, lineWidth=4)
drawLine(self.x + 30, self.y + self.height,
         self.x + 30, self.y + self.height + 12, lineWidth=4)
```

Obstacles: Drawing



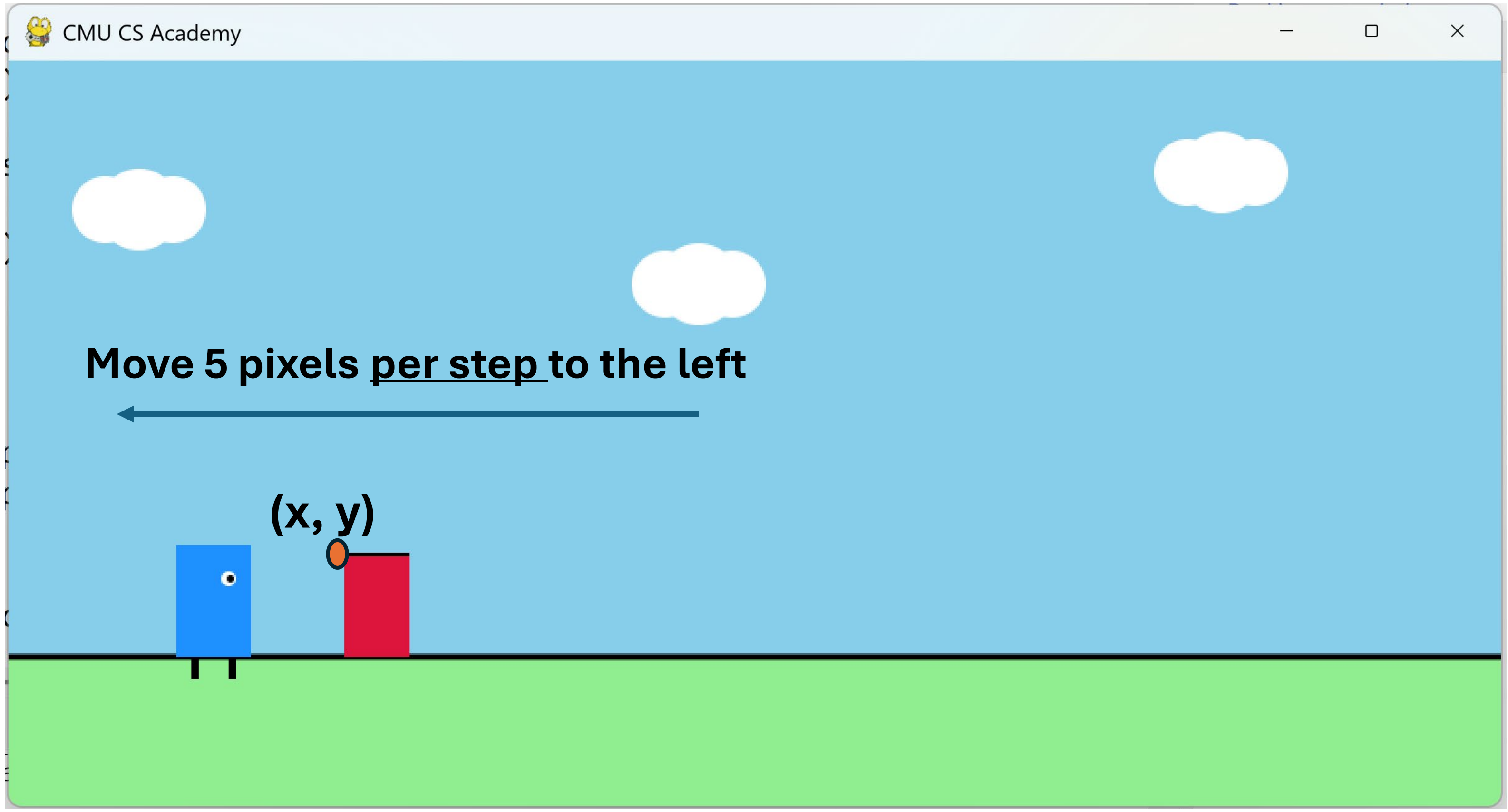
Obstacle width chosen randomly from [25,30,35]
Obstacle height chosen randomly [35, 45, 55]

Announcements

- Homework 8 due today
  - We already did the Cart and WishList question ☺
- Mentor Meetings focus on Term Project
- Term Project:
  - Do not include cmu_graphics in your submission on Gradescope
    - Email me your submission without cmu_graphics
    - We need something uploaded on Gradescope to grade your submission and give you feedback
  - Deductions for now showing Steady progress on GitHub will be applied on TP2 and TP3
    - Go to Office Hours if you still have issues
- Gradebook is updated with homeworks and the current Quiz Average (not attendance yet)
- Assessments Decisions will be announced later

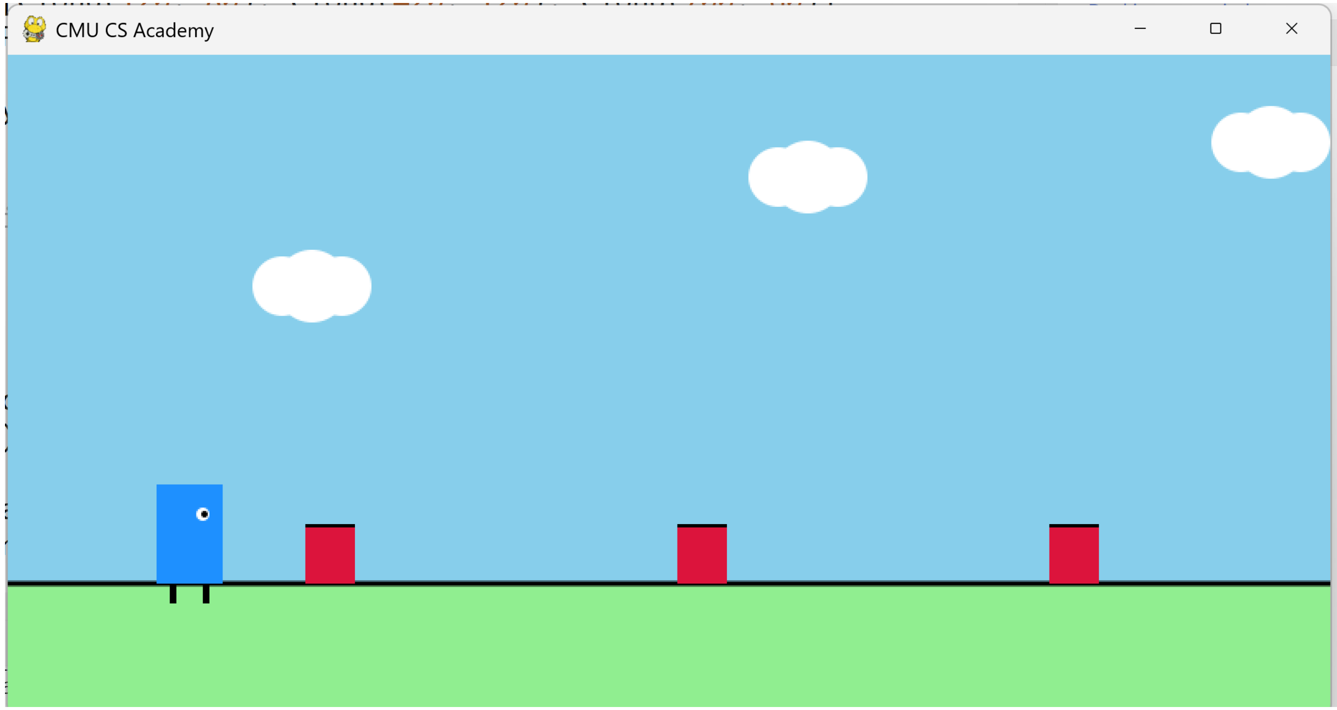
Obstacles: Motion

- Define a scrolling speed
  - Later: make speed higher to increase difficulty (next level)



Obstacles: Multiple

- I want to spawn with a defined time interval…
  - every x steps, add new obstacles
  - Making this time interval smaller, adds difficulty (next level)
  - New obstacle starts at the right of the screen

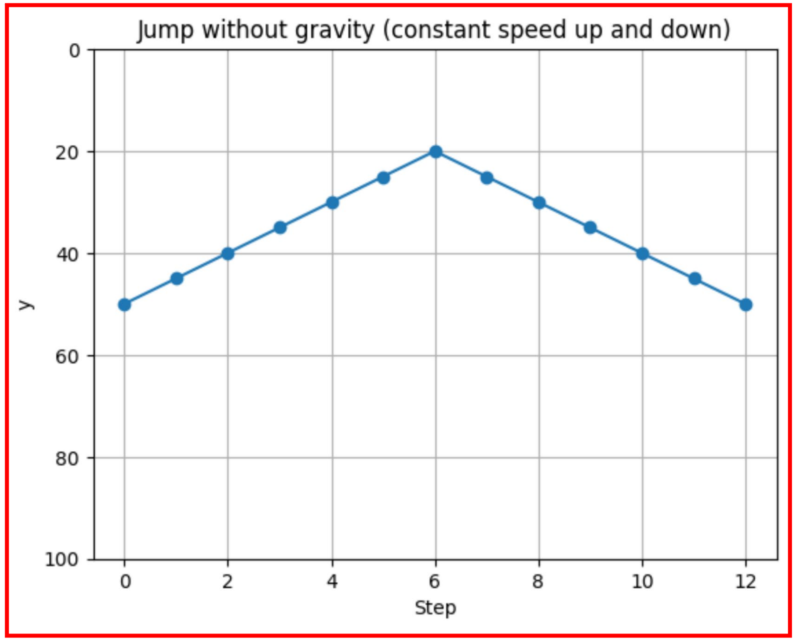


Player: Jumping

- To make something jump, we change its **y value over time**.
  - First, **y decreases** → the player goes **up**
  - Then it **stops, changing direction** at the top
  - Then **y increases** → the player comes **down**

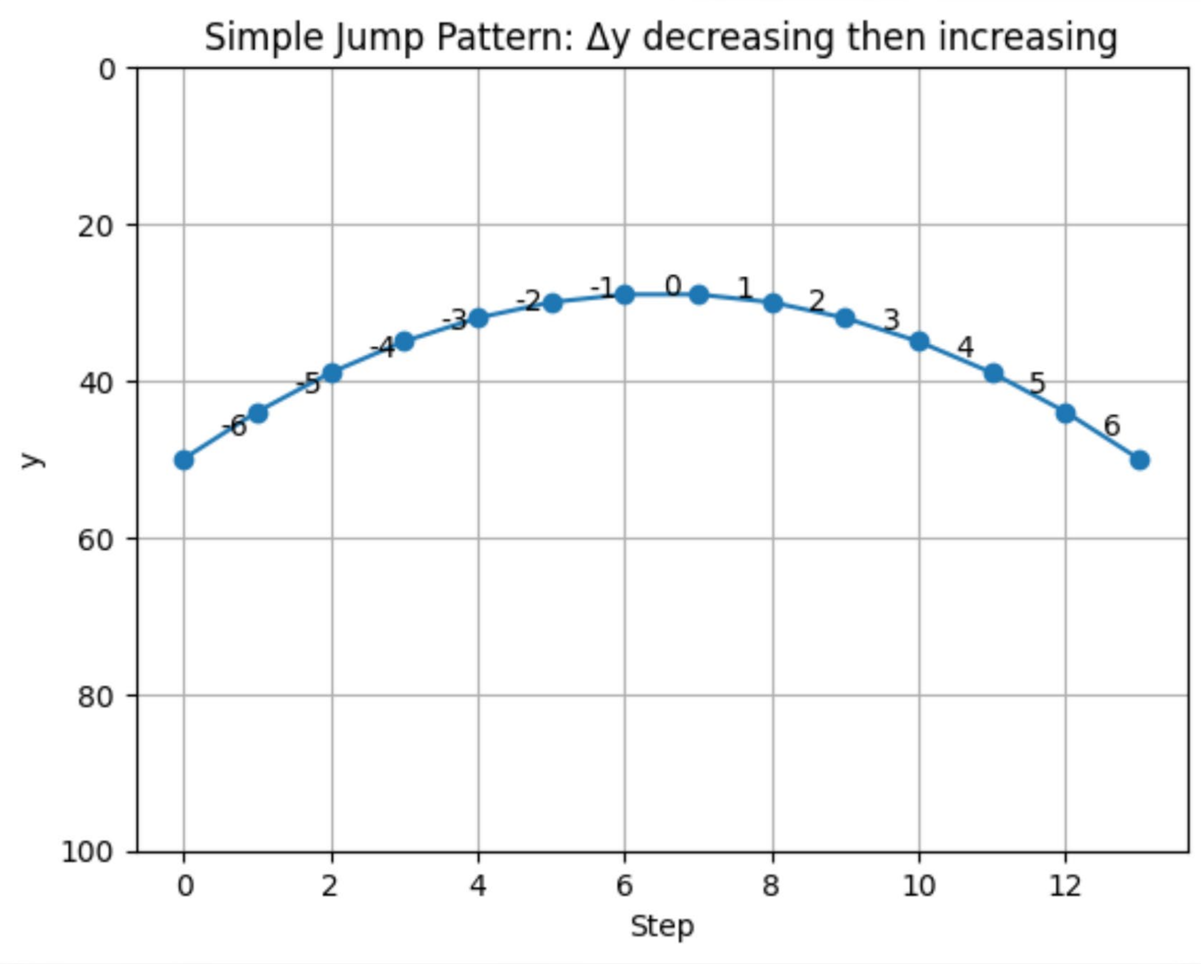
Without Gravity:
Player goes up and down at constant speed

(y changes by the same amount at every step)



Player: Jumping with Gravity

- A real jump is not just up then down — it changes speed. So instead of moving at a constant rate:
  - **Going up:**
    - y **decreases quickly at first**
    - then **slower and slower**
    - until it **stops at the top**
  - **Coming down:**
    - y starts increasing **slowly**
    - then **faster and faster**
    - until the player lands



- Start at y = 50
- Then change by:
  - -6, -5, -4, -3, -2, -1 → going up, slowing down
  - 0 → stop (top of jump)
  - +1, +2, +3, +4, +5, +6 → going down, speeding up

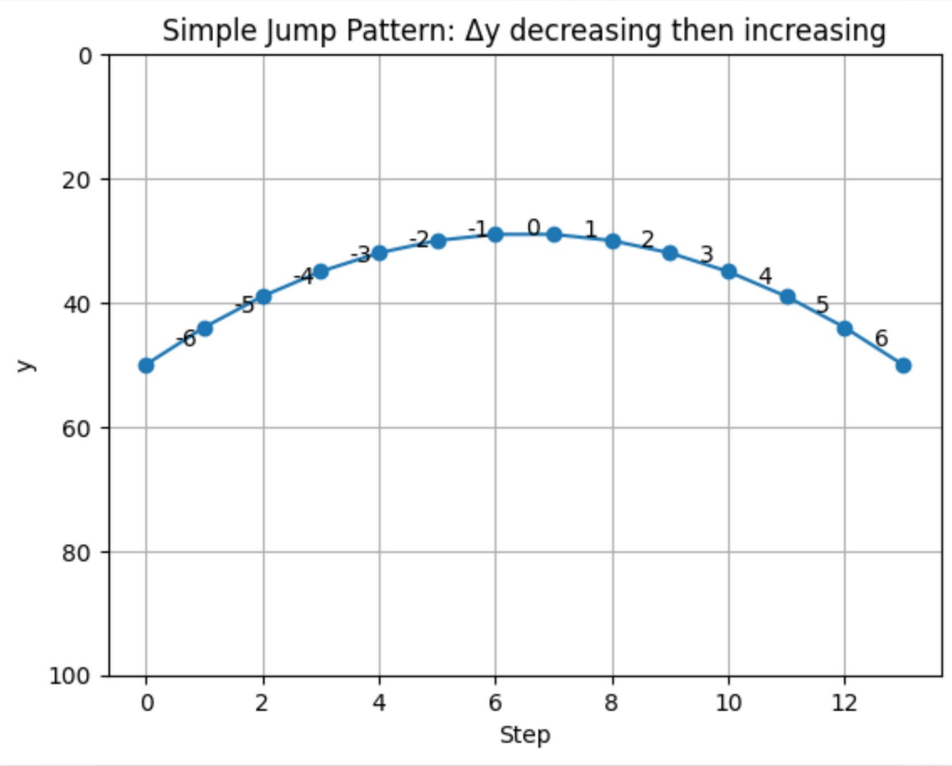
Player: Jumping with Gravity

<u>**In Coding:**</u>

- We will define a variable called (**dy**) which will define <u>how y changes</u> ….
- **dy** = {-6, -5, -4, -3, -2, -1, 0, 1, 2, 3, 4, 5, 6}
- It increases by (1) in each step
- 1 is the **gravity** effect

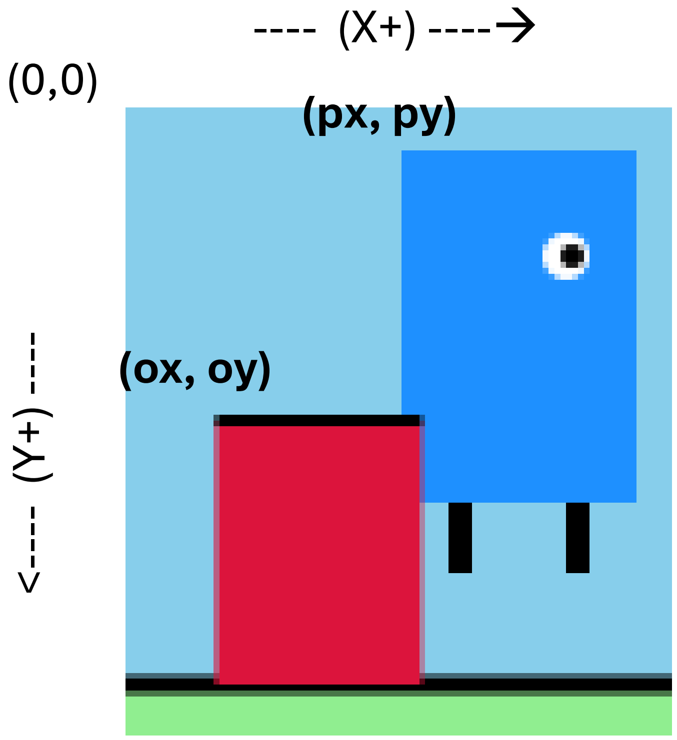
- gravity = 1
- dy = -6 (when space key is pressed)
- At every step or update:
  - y += dy
  - dy += gravity

- Start at y = 50
- Then change by:
  - -6, -5, -4, -3, -2, -1 → going up, slowing down
  - 0 → stop (top of jump)
  - +1, +2, +3, +4, +5, +6 → going down, speeding up



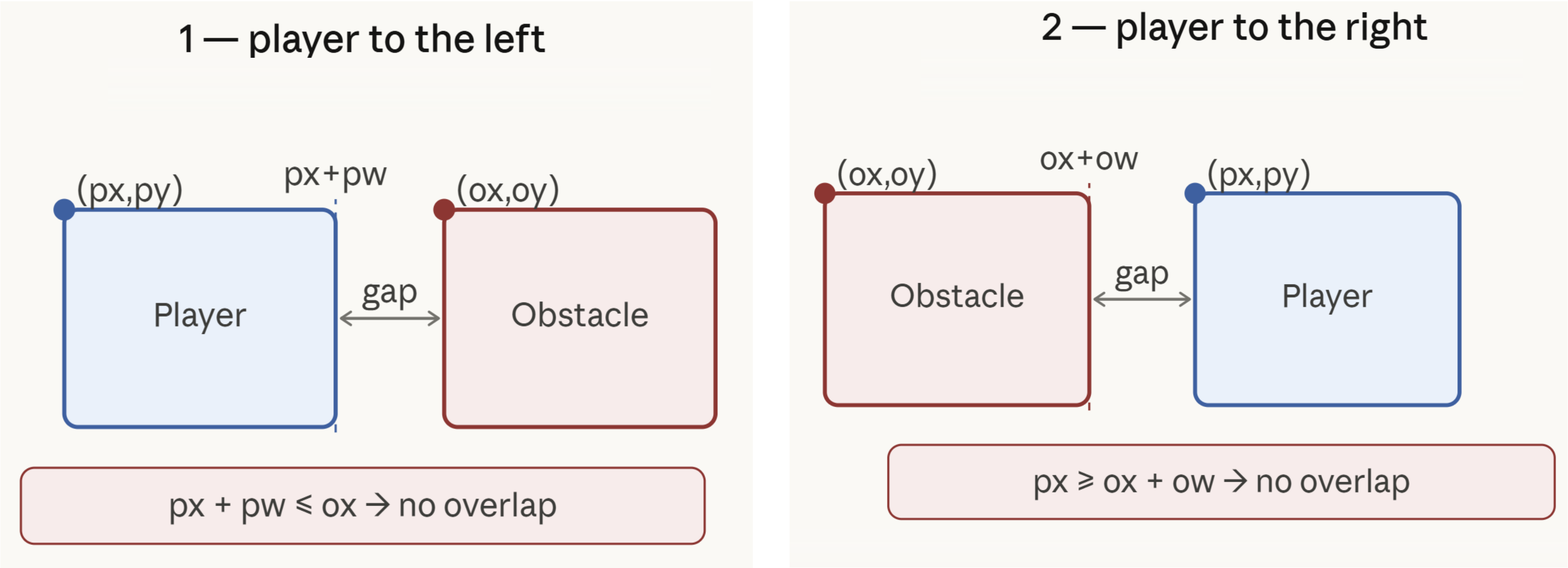
Detecting Collision

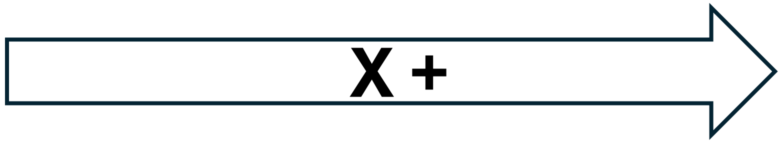
- Objects are Rectangles
- A collision happens if rectangles are not completely separated (Overlap)
- If they overlap in x **and** overlap in y → collision



No Collision: if they only overlap in one axis (y) or (x)

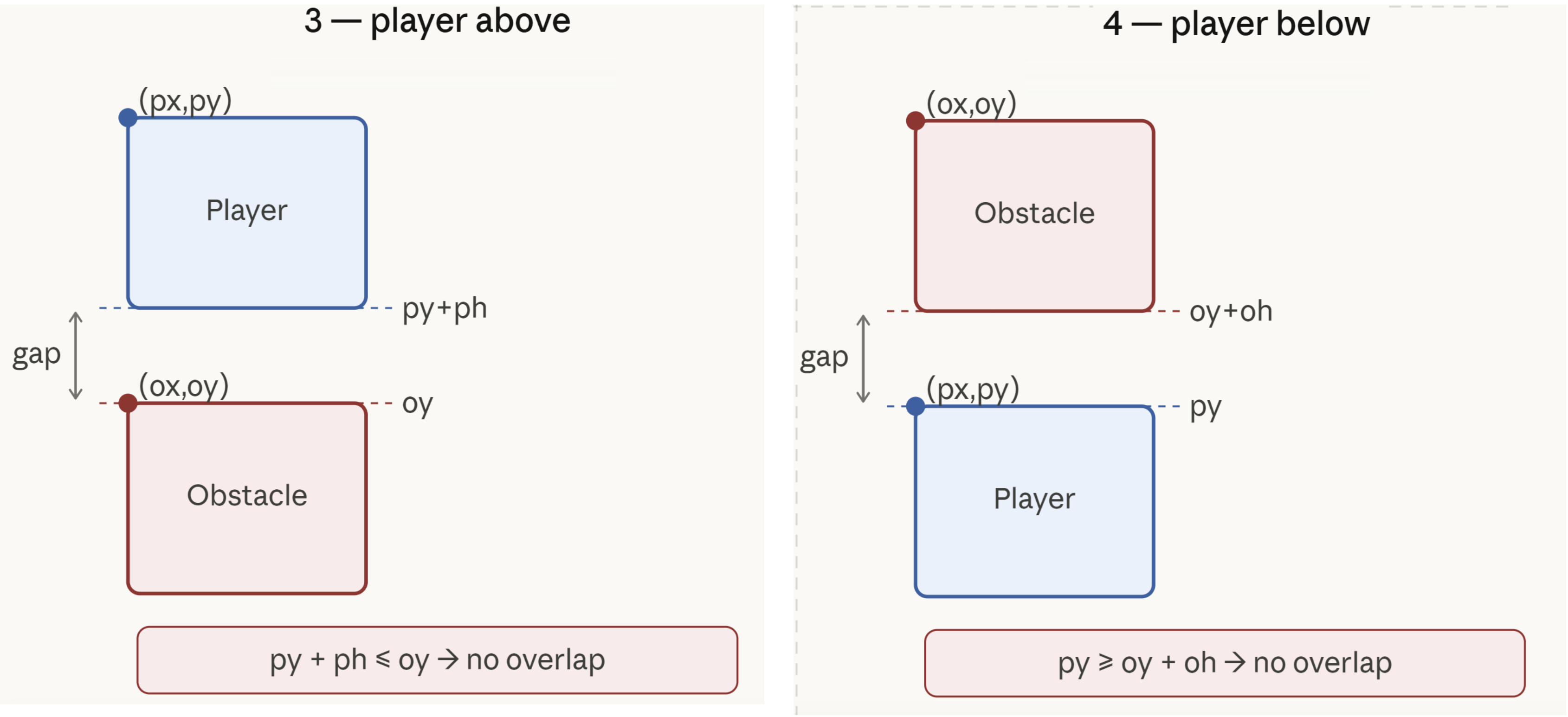
Overlapping
only vertically
(in the Y axis)

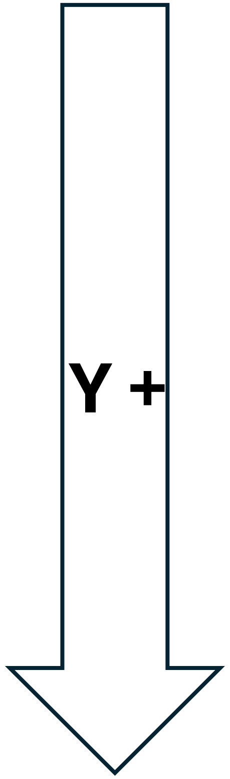




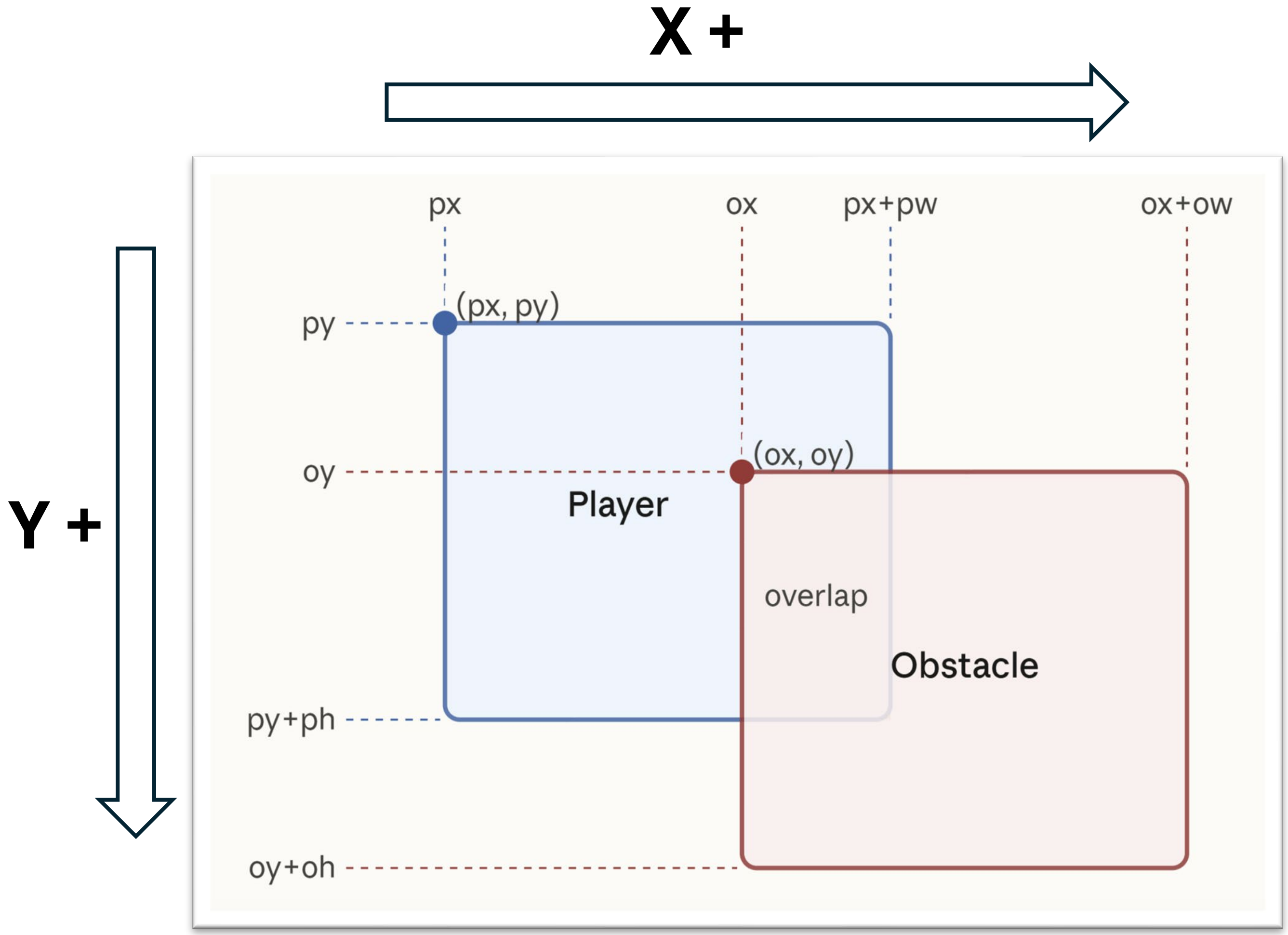
No Collision: if they only overlap in one axis (y) or (x)

Overlapping **only horizontally**
(in the X axis)





Detecting Collision: Overlapping in both axes



Overlap in X-axis

The player has entered the obstacle space from the left and hasn't slid fully past the obstacle to the right

- **px + pw > ox** — "the player's right edge has gone past the obstacle's left edge"
- **px < ox+ow** — "the player's left edge has not yet passed the obstacle's right edge"

Overlap in Y-axis

The player has entered the obstacle's space from above and hasn't fallen fully below the obstacle

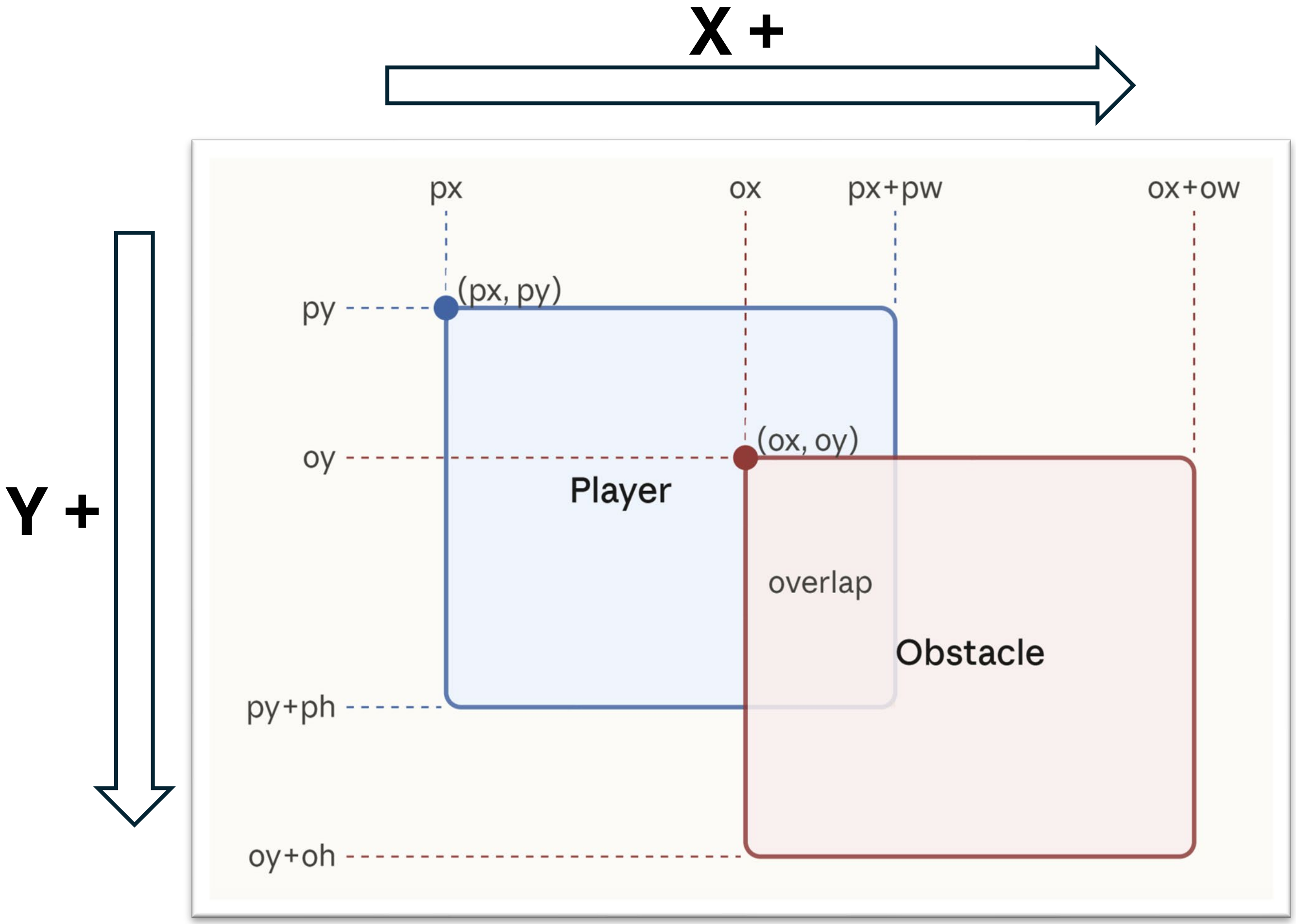
- **py + ph > oy** — "the player's bottom edge has gone past the obstacle's top edge"
- **py < oy + oh** — "the player's top edge has not yet passed the obstacle's bottom edge"

Detecting Collision: Overlapping in both axes

Variable	Value
`px`	140
`py`	100
`pw`	180
`ph`	160
`ox`	260
`oy`	160
`ow`	180
`oh`	160

`px+pw = 320`, `py+ph = 260`,

, `ox+ow = 440`, `oy+oh = 320`



Overlap in X-axis

px < ox+ow
140 < 440 ✓

px+pw > ox
320 > 260 ✓

Overlap in Y-axis

py < oy+oh
100 < 320 ✓

py+ph > oy
260 > 160 ✓

All four conditions are True → Collision